Complete gadget dichotomy for certificate-to-conjunction-DAG lifting

Yaroslav Alekseev, Yuval Filmus, Alexander V. Smal

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Editorial note. Actual complete journal article is CC BY4.0, DOI10.1007/s00037-025-00276-5; CCC abstract is identity alias only. Faithful editable transcription. Theorem4.3 counts a different computation-model classification. Full original Theorem3.4 projection proof and its leaf-only observation are included because the author invokes them verbatim for Theorem4.2. Lemma3.8 and complete conjunction-DAG definitions are retained. Source notation preserved; no author argument is supplied or repaired.

Allowed standard prerequisites (editorial).

- Conjunction DAG model. Garg et al.2020; exact author definition retained

Block composition (author source p.2)

For $f : \{0,1\}^n \to V$ and $g : \{0,1\}^m \to \{0,1\}$, a (block-)composition $f \diamond g : \{0,1\}^{n\times m} \to V$ is defined by

$$(f \diamond g)(z_1, z_2, \ldots, z_n) := f(g(z_1), g(z_2), \ldots, g(z_n)),$$

where each $z_i \in \{0,1\}^m$.

Prerequisites (author source pp.10–12)

All the logarithms are base 2. For $g : \{0,1\}^k \to \{0,1\}$, we define a function $g^n : \{0,1\}^{n\times k} \to \{0,1\}^n$ such that

$$g^n(x_1, x_2, \ldots, x_n) := (g(x_1), g(x_2), \ldots, g(x_n)).$$

For a function $f : \{0,1\}^n \to \{0,1\}$, an input $x \in \{0,1\}^n$, and a partial assignment $\sigma \in \{0,1,*\}^n$ that agrees with x on all non-$*$ coordinates, we use $f|_\sigma$ to denote a restriction of f to σ and $x|_\sigma$ to denote a projection of x to the $*$-coordinates of σ.

Decision tree complexity. A decision tree is a rooted binary tree with internal nodes labeled by input variables (representing queries), edges labeled with 0 and 1, and leaves labeled by some values. Decision tree evaluation is defined in a natural way: given an assignment for the input variables, we traverse the tree starting from the root and then sequentially choose every next edge according to the given assignments until we reach some leaf. The label of this leaf is the result of the evaluation. A decision tree computes some function f if for all possible assignments to the input variables, the decision tree evaluates to the value of the function. For a function f, $\mathrm{DT}(f)$ denotes the minimal depth of a decision tree computing f, and $\mathrm{DTSize}(f)$ denotes the minimal number of leaves in a decision tree computing f.

Certificate complexity. A certificate complexity is a non-deterministic analogue of the decision tree complexity. A certificate for an input $x \in \{0,1\}^n$ to a function f is a set $S \subseteq [n]$ of indices such that f restricted to all inputs that match x on S is constant, i.e., $f(y) = f(x)$ whenever $y|_S = x|_S$. The certificate complexity $\mathrm{C}(f,x)$ of f at input x is the size of the smallest certificate for x. Finally, the certificate complexity of function f is defined as $\mathrm{C}(f) := \max_{x\in\{0,1\}^n} \mathrm{C}(f,x)$.

3.1 Resistant gadgets (author source pp.14–15)

Definition 3.1. *A gadget $g : \{0,1\}^m \to \{0,1\}$ is resistant if for every $i \in [m]$ and $b \in \{0,1\}$, the function obtained by fixing the ith input to b is not constant. Equivalently, the minimum certificate complexity at inputs is larger than 1. E.g., the XOR function is resistant while the AND function is not.*

Theorem 3.2. *For any function $f : \{0,1\}^n \to \{0,1\}$ and a resistant gadget $g : \{0,1\}^m \to \{0,1\}$,*

$$\log \mathrm{DTSize}(f \diamond g) = \Omega(\mathrm{DT}(f)).$$

In the following proofs, we use $x_1, \ldots, x_n$ to refer to the inputs of function f. The inputs to the lifted function $f \diamond g$ are denoted by $z_{i,j}$, where $i \in [n]$ and $j \in [m]$.

Proof using random projections. Let T be a decision tree for $f \diamond g$ of size S. We apply the following random projection. For each $i \in [n]$, choose an index $j \in [m]$ at random, and set $z_{i,j}$ to a random value $b \in \{0,1\}$. Set the remaining variables $z_{i,k}$, $k \neq j$, to $x_i, \bar{x}_i, 0, 1$ in such a way that $g(z_{i,1}, \ldots, z_{i,m}) = x_i$. This is possible since g is resistant, as we show below. The result is a decision tree T' for f. To see that we can always construct such a projection, let $h(z)$ be a non-constant function, and choose two inputs w^0, w^1 such that $h(w^b) = b$. We define the projection ρ as follows: if $w^0_i = w^1_i = a$ then set $z_i = a$; otherwise, if $w^0_i = 0$ then set $z_i = x$, and if $w^1_i = 0$ then set $z_i = \bar{x}$. By construction, $\rho|_{x=b} = w^b$ and so $h|_\rho(b) = h(w^b) = b$. For example, suppose that $h(z_1, z_2, z_3)$ is the majority function on three inputs and that we set the first bit z_1 to 0. Then $h(0, 1, x_i) = x_i$ and $h(0, x_i, x_i) = x_i$ are two possible ways to choose a pair of inputs (w^0, w^1) corresponding to $(010, 011)$ and $(000, 011)$, respectively.

We would like to say that there is a choice of random projection for which T' is shallow. To do this, let us calculate the probability that some vertex v at depth D in T survives in T'. The path leading to v queries variables from at least D/m different blocks. For each such block, choose an arbitrary variable that is queried along the path and consider the value the path assigns to it. If the random projection sets this variable to the opposite value then the vertex v does not survive in T'. It happens with probability at least $1/(2m)$ for each variable, and so the vertex v survives with probability $p \leq (1 - 1/(2m))^{D/m} = 1/2^{O(D)}$. If $pS < 1$, then the expected number of vertices at depth D surviving in T' is less than 1, and so there is a random projection for which T' has a depth less than D. This happens for some $D = O(\log S)$. □

3.2 Weakly resistant gadgets (author source pp.16–17)

There are gadgets, such as $x \vee (y \wedge z)$, which are clearly not resistant, but for which the lifting still holds as we will show below. To capture these cases, we define a more general class of weakly resistant gadgets.

Definition 3.3. *A gadget $g : \{0,1\}^m \to \{0,1\}$ is weakly resistant if for every certificate α (a partial assignment which sets the value of g) there is a substitution of one input variable (with 0 or 1) which conflicts with α and does not make g constant.*

Every resistant gadget is trivially weakly resistant: we can take any variable mentioned in α and substitute the opposite value. The aforementioned gadget $h(x, y, z) = x \vee (y \wedge z)$ is weakly resistant (as we show below) but not resistant, since $h|_{x=1} \equiv 1$. The gadget $x \vee y$ is not even weakly resistant due to the certificate $x = y = 0$: if we substitute $x = 1$ or $y = 1$ then the gadget becomes constant. Let us verify that h is weakly resistant by considering all possible certificates:

- $x = 1$: take $x = 0$.
- $y = z = 1$: take $y = 0$ or $z = 0$.
- $x = y = 0$: take $y = 1$.
- $x = z = 0$: take $z = 1$.

We now prove the following lifting theorem:

Theorem 3.4. *For any $f : \{0,1\}^n \to \{0,1\}$ and a weakly resistant gadget $g : \{0,1\}^m \to \{0,1\}$ the following holds:*

$$\log \mathrm{DTSize}(f \diamond g) = \Omega(\mathrm{C}(f)).$$

Proof. Given a decision tree T for $f \diamond g$ of size S, we construct a blockwise random projection ρ in the following way. For each $i \in [n]$, we choose a random certificate C_i of g. Let $z_{i,j} = b$ be the substitution promised by weak resistance for C_i that does not make $g(z_i)$ constant. We can complete $\rho|_{z_i}$ to a projection

such that $g(\rho|_{z_i}) = x_i$ (as in the case of resistant predicates, see proof of Theorem 3.2 using random projections). Let ℓ be a leaf of T and τ be the corresponding partial assignment to the variables of $f \diamond g$. Suppose that τ contains certificates for at least D blocks (i.e., in these blocks, the value of g on τ is fixed). If there are M certificates of g, then τ is consistent with ρ with probability at most $(1-1/M)^D$. Choose $D = \Theta(\log S)$ so that $(1-1/M)^D S < 1$. By union bound, under some choice of ρ, the partial assignments for all leaves of $T|_\rho$ contain certificates for less than D blocks. Since by construction $(f \diamond g)|_\rho = f$, this shows that $\mathrm{C}(f) < D$. Indeed, given an input x to f, consider the corresponding input $\rho(x)$ to $f \diamond g$. This input reaches some leaf ℓ, in which less than D blocks contain certificates, say the blocks in $I \subseteq [n]$. For each input x' consistent with gadget values for blocks in I, we can find an input z reaching the same leaf such that $g^n(z) = x'$. Hence, the block certificates in ℓ correspond to a certificate of f. □

Remark. One of the anonymous reviewers noticed that the structure of a decision tree is completely irrelevant to the above proof. We only use the fact that the leaves are monochromatic and cover all the inputs. Thus, Theorem 3.4 can be restated by replacing the decision tree size with the subcube cover number. This also explains why, in the next section, we will be able to reuse the above proof in a similar lifting result for conjunction DAGs (see Theorem 4.2).

Corollary 3.5. *For any $f : \{0,1\}^n \to \{0,1\}$ and a weakly resistant gadget $g : \{0,1\}^m \to \{0,1\}$ the following holds:*

$$\log \mathrm{DTSize}(f \diamond g) = \Omega(\sqrt{\mathrm{DT}(f)}).$$

Proof. Beals et al. (2001) shows that for any function f,

$$\mathrm{DT}(f) \leq (\mathrm{C}(f))^2.$$

Together with Theorem 3.4, that gives us the desired bound. □

3.3 Gadget classification (author source pp.18–19)

Now, we can prove a dichotomy result. First of all, we need the following classification of gadgets.

Lemma 3.8. *For every $g : \{0,1\}^m \to \{0,1\}$, one of the following cases holds:*

(i) Function g is a (possibly empty) conjunction or a disjunction of literals.

(ii) Function g is weakly resistant.

Proof. Let $y_1, \ldots, y_m$ denote the inputs of g. If g is not weakly resistant then there is a certificate α such that for every substitution $y_j = b$ conflicting with α,

the function $g|_{y_j=b}$ is constant. Moreover, we can assume that α is an inclusion-minimal certificate. Suppose that $\alpha = \{y_{i_1} = b_1, \ldots, y_{i_t} = b_t\}$ and $g|_\alpha = b$. By assumption, the function $g|_{y_{i_s}=\bar{b}_s}$ is constant. Since α is inclusion-minimal, necessarily $g|_{y_{i_s}=\bar{b}_s} = \bar{b}$. Thus if $b = 1$ then

$$g = [y_{i_1} = b_1] \wedge \cdots \wedge [y_{i_t} = b_t],$$

otherwise, if $b = 0$ then

$$g = [y_{i_1} = b_1] \vee \cdots \vee [y_{i_t} = b_t].$$

□

We can now state the dichotomy theorem.

3.4 Generalization to search problems (author source p.19)

Let f be a relation, $f \subseteq \{0,1\}^n \times V$. For a gadget $g : \{0,1\}^m \to \{0,1\}$, we define a composition $f \diamond g \subseteq (\{0,1\}^m)^n \times V$ as a relation, such that

$$(z_1, z_2, \ldots, z_n, r) \in (f \diamond g) \iff (g(z_1), g(z_2), \ldots, g(z_n), r) \in f.$$

One can observe that the proofs of Theorem 3.2 and Theorem 3.4 work as well when we let f to be a relation. However, the Corollary 3.5 does not hold for the relations since $\mathrm{DT}(f)$ can be exponentially greater than $\mathrm{C}(f)$.

4.1 Conjunction DAGs (author source p.20)

A conjunction DAG was first defined formally in Garg et al. (2020), though it appears implicitly in previous work. A conjunction DAG over a set of variables is a single-rooted DAG with the following additional information:

- Each internal vertex is annotated with a variable, and it has two outgoing edges, one labeled 0 and the other labeled 1.
- Each vertex v is annotated with a partial assignment $\rho(v)$ with the following constraint. Suppose that v queries x_i, and the answer b leads to the vertex v_b. Then the partial assignment $\rho(v_b)$ is a subset of the partial assignment $\rho(v) \cup \{x_i \leftarrow b\}$. (This is not identical to the definition in Garg et al. (2020), but morally the same.)
- The partial assignment at the root is the empty assignment.
- Each leaf is annotated with some value.

A decision DAG computes f if for every leaf ℓ annotated with y_ℓ, $\rho(\ell)$ is a y_ℓ-certificate of f. Every decision tree is a decision DAG. There are three parameters of interest: the (total) size (number of vertices), the leaf size (number of leaves), and the width (maximum number of variables in any $\rho(v)$).

4.4 Gadget classification (author source pp.22–23)

Moreover, for conjunction DAGs we can prove an analog of Theorem 3.4:

Theorem 4.2. *Let $f \subseteq \{0,1\} \times V$ be a relation and $g : \{0,1\}^m \to \{0,1\}$ be a weakly resistant gadget. If there is a conjunction DAG for $f \diamond g$ with S leaves, then the certificate complexity of f is at most $O(\log S)$.*

Proof. The proof of Theorem 3.4 can be applied in this setting as is because it ignores the internal structure of the decision tree and considers only the leaves. □

Note that Theorem 4.2 gives us a lifting from certificate complexity to leaf size. Unfortunately, in the case of the falsified clause problem for CNF, this theorem cannot be effectively used to prove lower bounds since leaf size, as well as certificate complexity, is usually small. However, this theorem still implies a dichotomy result similar to one in Theorem 3.9.

Theorem 4.3. *For every gadget $g : \{0,1\}^m \to \{0,1\}$, one of the following cases holds:*

(i) There is an infinite family of functions f_n such that $f_n \diamond g$ can be computed by a conjunction DAG having $O(n)$ leaves, but $\mathrm{C}(f_n) = \Omega(n)$.

(ii) For every relation f, if we have a conjunction DAG for $f \diamond g$ with S leaves, then the certificate complexity of f is at most $O(\log S)$.

Proof. The two cases of the theorem correspond to the two cases of Lemma 3.8. If g is a disjunction then we take $f_n := \mathrm{OR}_n$. Clearly, $\mathrm{C}(f_n) = n$ and $f_n \diamond g$ can be computed with a conjunction DAG (actually a tree) having $O(n)$ leaves (if g is constant, then the DAG has only one leaf). The case of a conjunction is similar. Otherwise, gadget g is weakly resistant and by Theorem 4.2 we have $\log S = \Omega(\mathrm{C}(f))$. □

References

Source reference used as standard background

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